



Technical Service Bulletin: TSB170045

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Crankcase Pressure Sensor Fault Code 555 or 556: Reoccurring Issue in Cold Weather

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX CM871
- ISX CM871 E
- ISX15 CM2250
- ISX15 CM2250 SN
- ISX15 CM2350 X101
- PowerGen QSX15 CM2250
- QSX15 CM2250 ECF
- QSX15 CM2250 X115
- QSX15 CM2350 X105
- QSX15 CM2350 X106
- X15 CM2350 X114B
- X15 CM2350 X116B

Issue

Symptom:

- Repeat occurrence of high crankcase pressure Fault Codes 555 or 556 immediately after engine start up at ambient temperatures below freezing.

Root Cause:

- Excessive moisture can lead to ice build-up inside crankcase pressure sensor. Ice presses against crankcase pressure sensor diaphragm and produces a false occurrence fault code.

Verification

The following should be addressed as an additional step when inspecting the crankcase pressure sensor as detailed in the troubleshooting process for a FC555 or FC556. For accurate analysis, complete the following inspection as close to the fault code detection as possible.

- Verify engine has been operating in temperatures below freezing
- Locate crankcase pressure sensor; on inlet hose to crankcase breather or on crankcase breather.
- If crankcase pressure sensor measures within range as stated in Troubleshooting step 2A - Verify Pressure Sensor Accuracy, remove crankcase pressure sensor and inspect for evidence of water accumulation or ice in crankcase pressure sensor port as seen Figure 1 below. See corresponding service manual. Reference Procedure 019-445 in section 19.



Figure 1, Frozen Crankcase Pressure Sensor.

Resolution

- For ISX15 CM2250, ISX15 CM2250 SN, and ISX15 CM2350 X101 engines a crankcase breather retrofit kit is available. Follow Installation Instructions 5414650 to replace crankcase breather assembly with improved crankcase breather.
- Coating crankcase pressure sensor with CRC 3-36 or WD-40 prevents water or debris accumulation in crankcase pressure sensor port that may freeze when temperatures drop below freezing. See Service Instruction section below.
- Periodic reapplication of CRC 3-36 or WD-40 spray in cold weather as a preventative measure is recommended.

Continual Preventive Maintenance Without Retrofit: Service Instructions

- Let crankcase pressure sensor thaw if frozen debris is found. Spray CRC 3-36 or WD-40 directly into the crankcase pressure sensor port. Do **not** insert nozzles, straws, or other attachments into the crankcase pressure sensor port or damage can occur.
 - Verify crankcase pressure sensor port receives a complete coating of CRC 3-36 or WD-40 spray.

- Install crankcase pressure sensor. See corresponding Service Manual. Reference Procedure 019-445 in Section 19.
- Start and run engine for 5 minutes to verify fault codes are inactive and crankcase pressure sensor is properly functioning.
- Continue with troubleshooting steps for FC555 or FC556.

Document History

Date	Details
2017-4-7	Module Created
2017-9-12	Removed ISX CM570, ISX CM870, and QSX15 CM570.
2018-7-11	Added new retrofit kit to Resolution section.

Last Modified: 08-Aug-2018
